

during the post period for the years the data are available.

A second limitation is that our data do not include information on the cause of death. The death information in the Census Numident is derived primarily from the Social Security Administration death records, which contain only date of death. We therefore supplement our main analysis with data from the 2008 year of the ACS, which was linked to death certificate records from 2008 to 2015 as part of the Mortality Disparities in American Communities (MDAC) project. While this drastically reduces both the sample size and follow-up period, it does allow us to conduct exploratory analyses of changes in mortality based on the underlying cause of death as reported on the death certificate.

IV Empirical Strategy

Our empirical strategy looks at changes in annual mortality in the expansion states relative to the non-expansion states before and after the implementation of the ACA Medicaid expansions. We estimate this using an event-study model that allows us to assess the evolution of relative outcomes while controlling for fixed differences across states and national trends over time. We estimate:

$$Y_{isjt} = Expansion_s \times \sum_{\substack{y=-6 \\ y \neq -1}}^3 \beta_y I(t - t_s^* = y) + \beta_t + \beta_s + \beta_j + \gamma \mathbf{I}(j = t) + \epsilon_{isjt}. \quad (1)$$

As described earlier, our data is constructed at the individual (i) by year (t) level. Each individual responds to the ACS during a survey wave (j) and reports their state of residence (s). In our first stage analysis, Y_{isjt} measures eligibility or coverage in each year. For our analysis of mortality, the dependent variable is $Died_{isjt}$, which denotes death during each year t among individuals who were alive at the beginning of year t . We only observe mortality over a partial year during the year of the individual's ACS interview (j), since that individual had to be alive in order to respond to the survey. To account for this, we include an indicator variable that year t is the year that the individual responded to the ACS (i.e., that $j = t$).¹² In this equation, β_s denotes state fixed effects and β_j denotes fixed effects associated with each survey wave. β_t denotes calendar year fixed effects, which account for general trends in mortality for all individuals in our sample, including their gradual aging over time.

The variable $Expansion_s$ equals 1 if individual i was living in a state that opted to expand Medicaid eligibility between 2014 and 2017, and zero otherwise. Indicator variables $I(t - t_s^* = y)$ measure the time relative to the implementation year, t_s^* , of the expansion in each state, and are zero in all periods for non-expansion states.¹³ While most states expanded in the beginning of 2014, some states expanded later in the year or in subsequent years. If a state expanded on or after July 1 of a given year, we code it as having expanded in the subsequent year.¹⁴ The omitted category is $y = -1$, the year prior to the expansion. Therefore, each estimate of β_y provides the change in outcomes in expansion states relative

¹²We do not have information on the date of the ACS interview. If we drop the observations for which we observe less than a full year of mortality, our results are unchanged.

¹³We group together $y \leq -6$ into a single indicator variable interacted with expansion status since we only observe $y < -6$ for late expander states.

¹⁴In our analyses, states that expanded Medicaid in 2014 are AR, AZ, CA, CO, CT, HI, IL, IA, KY, MD, MI, MN, NJ, NM, NV, ND, OH, OR, RI, WA, and WV. Michigan implemented their expansion in April 2014 with the remainder of states expanding in January 2014. States that we considered to have 2015 expansions are NH (implemented August 15, 2014), PA (January 1, 2015), and IN (February 1, 2015). We consider AK (September 1, 2015) and MT (January 1, 2016) to be 2016 expansion states and LA (July 1, 2016) to be a 2017 expansion state.

to non-expansion states during year y , as measured from the year immediately prior to expansion. If mortality rates for expansion and non-expansion states were trending similarly prior to the ACA, we expect that estimated coefficients associated with event times $y = -6$ to $y = -2$ will be small and not statistically significant. We estimate equation (1) with a linear probability model and report heteroskedasticity-robust standard errors that are clustered at the state level.¹⁵ All analyses use ACS survey weights.

In addition to the event study analyses, we also present difference-in-differences (DD) estimates as a summary of the effect across all post-expansion years. These are estimated using the same equation except that the event study indicators are replaced with a single variable denoting an expansion state during the post period ($Expansion_s \times Post_t$). This indicator turns on starting in the year of expansion for each state.

Because we have a fixed sample that ages in each period, mortality rates increase over time (i.e., our sample is oldest in the last year, 2017). In this way, our analysis tracks the mortality trajectory for a fixed cohort defined as adults ages 55 to 64 in 2014 and is representative of the outcomes over time for this group. Since mortality rates are higher during the post-period for our sample due to aging, regardless of exposure to Medicaid, comparing our estimates of the effects of the ACA Medicaid expansions to the mortality rate in the pre-ACA period as a counterfactual is incorrect. Instead, we estimate a “counterfactual” rate for those living in expansion states that depends on average mortality rates observed during the post period. We calculate this rate as the average annual mortality rate in the expansion states in the post-expansion period minus the reduced form mortality difference-in-differences estimate. This rate tells us what the mortality rate would have been in the treated states in our sample if their mortality rates had followed the same trajectory we observe in the control states. By constructing the counterfactual rate in this manner, we are able to take into account both the aging of the sample and the mortality reduction of expanded Medicaid coverage during the post-period, see discussion in [Goodman-Bacon \(2016\)](#).

V Results

V.A Medicaid Eligibility, Medicaid Enrollment, and Insurance Coverage

We first estimate the impact of the ACA Medicaid expansions on Medicaid eligibility and coverage for individuals in our sample. We show that, using a variety of data sources and approaches, respondents residing in the expansion states experienced significant increases in Medicaid eligibility, Medicaid enrollment, and health insurance coverage relative to those in the non-expansion states. The effects we document among our disadvantaged sample are substantially larger than those estimated in the population overall (e.g. [Black et al., 2019](#)), indicating that the individual-level characteristics we used to define our sample are successful in identifying those most likely to be affected by the eligibility expansions.

¹⁵In Appendix Table A2, we examine the sensitivity of our results to using both a standard logistic regression and a Cox proportional-hazards model. The difference-in-differences estimates from these nonlinear models are modestly smaller in magnitude than the linear probability model, but continue to show large and statistically significant reductions in mortality as a result of the ACA Medicaid expansions. In addition, our inference is similar if we cluster the errors at the Census division level or allow for spatial correlation of the errors using the methods proposed by [Conley \(1999\)](#). See discussion in Appendix Section 4 and results in Appendix Table A8.

First, we examine how eligibility for Medicaid changed in expansion relative to non-expansion states. Individuals eligible for Medicaid are “conditionally covered” by the program, in the sense that they may choose to remain uninsured and enroll only when they become ill. This concept of conditional coverage was first discussed by [Cutler and Gruber \(1996\)](#) in their study of historic Medicaid expansions for pregnant women and children; it may be even more relevant in our context, however, given another change under the ACA designed to make it easier for the uninsured to gain immediate access to Medicaid-funded services. For the first time, the federal government required states to allow presumptive eligibility under their Medicaid programs. Specifically, the ACA granted hospitals the ability to make presumptive eligibility determinations for Medicaid for certain groups covered in their state, including the non-elderly ACA expansion population ([Caucci, 2014](#)).¹⁶ This means that if patients appear to have incomes low enough to qualify for Medicaid, hospitals may grant temporary Medicaid enrollment. Recipients of this temporary enrollment status may immediately receive health services and providers are guaranteed reimbursement for those services. In addition to presumptive eligibility, federal law directs states to provide retroactive coverage for new enrollees by covering medical bills incurred up to 3 months prior to their application date if they met the eligibility criteria during that time.¹⁷ By not requiring an individual to first enroll in Medicaid prior to receiving Medicaid-funded care, these policies reinforce the notion that all eligible individuals are effectively covered by the program even if not actually enrolled.

We estimate Medicaid eligibility for our sample using information on state eligibility rules (see Appendix Section 2 for additional information) and characteristics of respondents in the 2008 to 2017 waves of the ACS who were ages 55 to 64 in 2014, and otherwise meet the same sample definition as used in our main analyses. Note that we are unable to use individual panel data for this analysis since we only observe respondent characteristics for our linked ACS-mortality sample during the year they completed the ACS survey. While repeated cross-sectional data for this cohort does not exactly mirror the individual panel data used to study mortality, it allows us to provide an estimate of the changes in eligibility likely similar to those experienced by our main sample. We rely on the same data to study changes in overall insurance coverage over time, since this information is also not captured in our panel data.

Next, we use the longitudinal administrative data on Medicaid enrollment to examine changes in Medicaid coverage in each year, and in the total number of years of Medicaid coverage experienced.¹⁸ We use three different dependent variables to measure coverage changes. The first is equal to 1 if the individual is enrolled in Medicaid during year t and 0 otherwise, capturing changes in coverage in each year. The second variable equals the number of days enrolled by the individual in each year, including zero days for those who do not enroll. Third, we examine the total days during which the individual was enrolled up to and including year t , divided by 365; i.e., the total accumulated number of years

¹⁶Previously presumptive eligibility programs were optional for states and limited to pregnant women and children. States also had discretion over what types of providers could grant presumptive eligibility for these groups.

¹⁷A handful of states (AR, IA, IN, NH) had federal waivers to waive retroactive coverage for the expansion population, or other existing Medicaid eligibility groups, during our study period ([Musumeci and Rudowitz, 2017](#)).

¹⁸In this analysis, we use Medicaid enrollment data for all individuals starting with the year of their ACS interview. However, since we have complete enrollment information for each year, we exclude the $\gamma\mathbf{I}(j = t)$ term from equation (1). This term is also excluded from the other first-stage analyses described above that rely on repeated cross-sections of the ACS.

of Medicaid enrollment. This third version acknowledges that health insurance coverage may have beneficial health effects that extend even after the period of coverage is over and that health benefits may accumulate over time.¹⁹

Finally, we also document how overall insurance coverage evolved following the expansion. In contrast to Medicaid enrollment, we do not have administrative data on insurance coverage. Instead, we use a repeated cross-section of respondents from the 2008 to 2017 waves of the ACS to measure point-in-time insurance coverage for individuals who meet our sample criteria, as we did to determine changes in eligibility.²⁰ We also explore the impact of the expansion on insurance coverage using two alternative data sources. First, we examine changes in contemporaneous insurance coverage using the National Health Interview Survey (NHIS), which is considered to have the most valid coverage estimates nationally (Lynch et al., 2011). In contrast to the ACS, the NHIS uses state-specific names for Medicaid/CHIP in its coverage questions; it also includes a verification question for the uninsured. Second, we use the panel Health and Retirement Study (HRS) to examine cumulative changes in insurance coverage. We use restricted versions of both datasets that include state identifiers and construct analytic samples using the same sample criteria as our main sample.

The results are presented in Figure I and in the first five columns of Table I. We find a large increase in Medicaid eligibility associated with the ACA Medicaid expansions with gains of between 49 and 51 percentage points during each post-expansion year, as compared to the year just prior to expansion. Consistent with many other studies of this policy, we find significant increases in Medicaid coverage and decreases in uninsurance associated with the decision to expand Medicaid eligibility. We find the probability an individual is enrolled in Medicaid during the year increases by 12.8 percentage points, and that on average, individuals in expansion states experience 43 additional Medicaid enrolled days per year relative to those in non-expansion states (with those who do not enroll coded as having 0 enrolled days). We aggregate the number of days enrolled in each year to examine the cumulative number of Medicaid enrolled years. Cumulative Medicaid enrollment experienced by our sample also increases significantly, with respondents in Medicaid expansion states experiencing 0.38 additional years of Medicaid relative to those in non-expansion states on average, or about 0.67 additional years by the end of our sample period.

Using data from the ACS, we find that self-reported uninsurance decreases by 4.4 percentage points, on average, following the expansions. The estimates for years 1 and 2 are larger than those for year 3, which likely reflects the increasing share of the sample that is aging into Medicare over the study period. Note that we find somewhat larger impacts of the Medicaid expansions on coverage using the NHIS, with decreases ranging from 4.9 to 9.5 percentage points across the post-expansion years, and an average decline in the post-expansion period of 5.8 percentage points (see Appendix Figure A1 and Appendix Table A3). Estimates from both the NHIS and ACS indicate that a significant share of new Medicaid enrollees would have some form of insurance coverage in the absence of Medicaid expansion. Some of this “crowd-out” may reflect the availability of subsidized exchange coverage for individuals

¹⁹See, e.g., Boudreaux et al. (2016); Brown et al. (2018); Currie et al. (2008); Goodman-Bacon (2016); Thompson (2017); Wherry and Meyer (2016); Wherry et al. (2017) for evidence of such long-term effects.

²⁰To assess comparability with our estimates based on administrative data, we also provide estimates of how self-reported Medicaid enrollment changed using the same data, although it is important to note that there is a well-known under-report of Medicaid enrollment in survey data (Boudreaux et al., 2015). See Appendix Figure A1 and Appendix Table A3.